

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why software tests fail and hardware attestation wins

arxiv.org/abs/2504.04715

The Model Substitution Problem

Economic incentives drive covert swaps in commercial LLM APIs

Core insight: Users pay for specific models (e.g., Llama-3-70B), but black-box APIs offer no verifiable guarantee the advertised model is actually being served. Providers face strong incentives to substitute cheaper alternatives while maintaining pricing — a classic information asymmetry problem.

Hypothesis Testing Framework

H_0 : API outputs come from the specified model $P_{\text{spec}}(y|x)$ | H_1 : API outputs come from an alternative $P_{\text{actual}}(y|x)$

Four Substitution Types



Smaller Models

Serve a smaller parameter count (e.g., 8B instead of 70B)

Dramatic cost reduction with plausible deniability



Quantized Variants

FP8 or INT8 instead of FP16 — nearly identical outputs

2× memory savings, minimal benchmark degradation



Fine-Tuned Versions

Updated or instruction-tuned variants of the same base model

Hard to distinguish without weight-level access



Different Families

Entirely different model architectures (e.g., Qwen for Llama)

Most aggressive substitution, highest cost savings

FP8 Quantization Preserves Most Accuracy

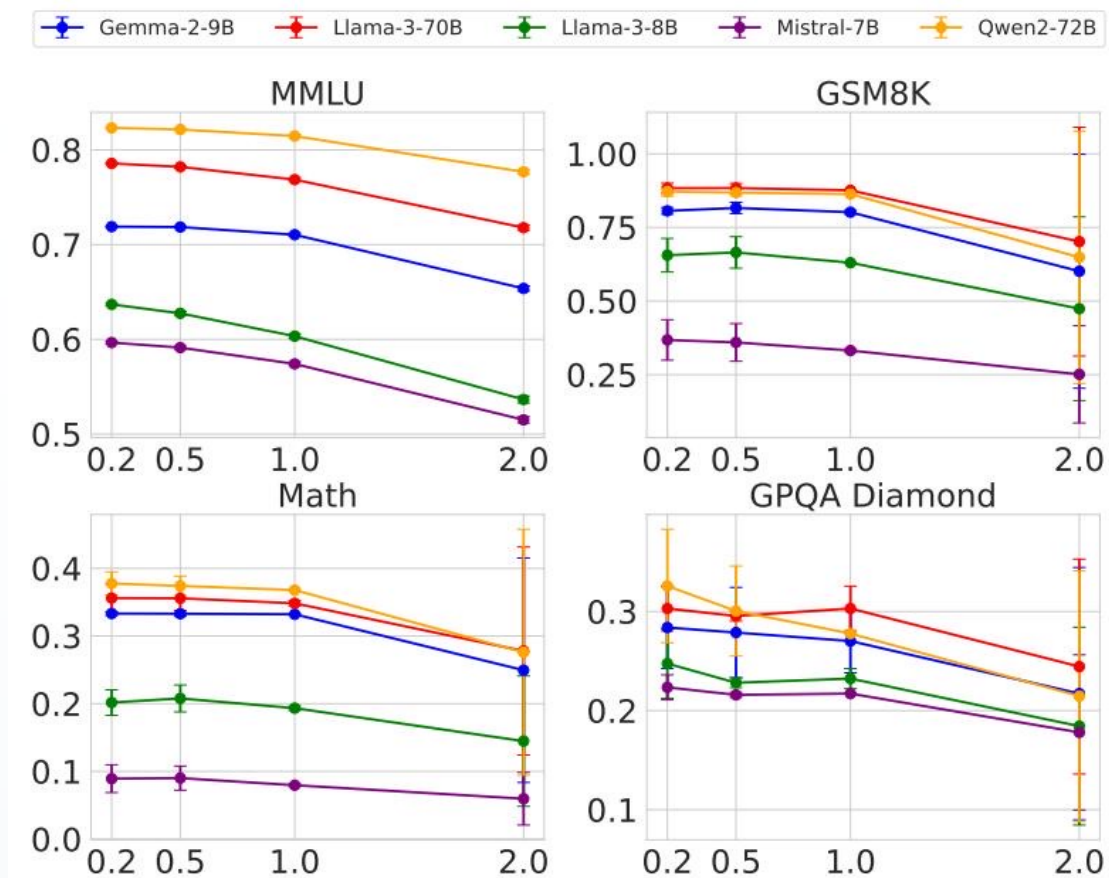
Making substitution extremely hard to detect via benchmark comparison

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

 **Key insight:**For most models, FP8 scores fall within the error bars of the original — especially on MMLU and GSM8K. The substitution signal is buried in statistical noise.

Higher Temperature Amplifies Gaps but Increases Noise

At T=2.0, performance degrades across all models — further obscuring the substitution signal



Temperature Scaling Effect

Experiments at T = 0.2, 0.5, 1.0, 2.0 show higher temperatures degrade all models' performance, not just substitutes.

Signal Obscuration

The gap between original and substitute narrows or becomes noisy at high T. Statistical separation becomes infeasible.

Realistic Deployment Setting

Temperature scaling is not an artificial test condition — it is standard in production APIs. Auditors must contend with this noise.

Log Probability Fingerprinting Fails

Production-level inference nondeterminism defeats metadata-based detection

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20



Hardware Variation

Same Llama-3-8B produces different log-probabilities on A100 vs H100 GPUs.
Hardware fingerprinting is unreliable.



Software Framework Drift

transformers 4.40 vs 4.50 and vLLM
0.6.2 vs 0.8.2 yield divergent token-level probabilities for identical inputs.



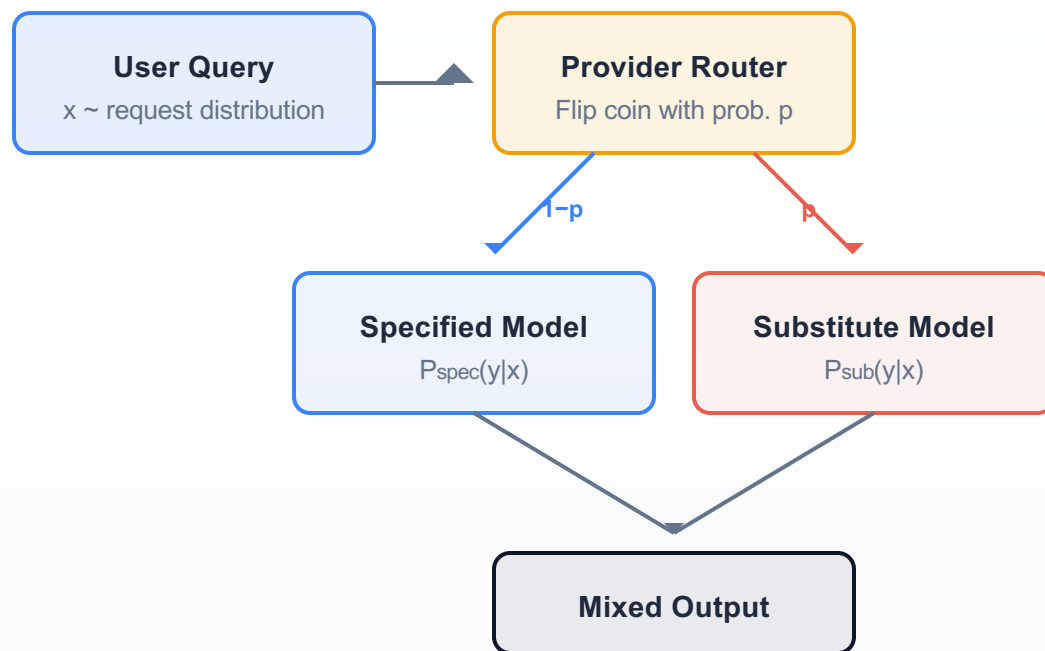
The Fundamental Problem

An auditor cannot distinguish between:
(a) legitimate model on different infra
(b) an actual model substitution

Randomized Routing Makes Detection Query-Prohibitive

As substitution probability p approaches 0, the mixed distribution converges to the specified model

Randomized Substitution Attack Model



$$D_{\text{mixed}} = (1-p) \cdot D_{\text{spec}} + p \cdot D_{\text{sub}} \rightarrow D_{\text{spec}} \text{ as } p \rightarrow 0$$



Query Complexity Explodes

As the substitution rate p decreases, the number of queries required for statistical detection grows exponentially. At $p = 0.01$, millions of API calls are needed — economically infeasible for auditors.



Benchmark Evasion

Providers can detect known benchmark prompts and route them to the genuine model while serving substitutes for regular traffic. This makes even benchmark-based auditing fundamentally unreliable.

Software-Only Methods Are Fundamentally Unreliable

Three failure modes demonstrate why software detection cannot close the trust gap



Text-Based Statistical Tests

e.g., Maximum Mean Discrepancy (MMD)

Weakness

Require impractically large query budgets to achieve statistical significance.

Practical Limitation

Fail against subtle or randomized substitutions where the mixed distribution converges to H_0 .

Query-heavy



Log-Probability Methods

Token-level probability fingerprinting

Weakness

Defeated by production-level inference nondeterminism across software and hardware stacks.

Practical Limitation

Different transformers versions and GPU generations (A100 vs H100) produce divergent fingerprints.

Unstable



Adversarial Countermeasures

Benchmark evasion + randomized routing

Weakness

Providers can actively detect audit patterns and route known benchmark prompts to the genuine model.

Practical Limitation

Randomized routing at low p makes statistical separation query-prohibitive. The adversary controls the API.

Active attack

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

How TEEs Guarantee Integrity

Hardware-Secured Enclaves

CPU/GPU memory regions isolated from the host OS and hypervisor

Cryptographic Attestation

Remote attestation proves exactly which code and data are loaded

Model Weight Verification

Attestation includes hash of model weights — proves correct model is loaded

Modest Performance Overhead

Typically 5–15% throughput reduction — acceptable for security-critical APIs

Practical Deployment Path

- NVIDIA Confidential Computing on H100 GPUs
- AMD SEV-SNP for CPU-based inference pipelines
- Cloud providers (AWS, Azure, GCP) already offer TEE instances
- Deployable today — unlike ZKPs, which are computationally prohibitive

for large transformer models.

TEEs vs Software-Only Detection

TEEs : Provable · Efficient · Robust · Deployable now

Software : Probabilistic · Expensive · Fragile · Adversarially defeated

Unlike zero-knowledge proofs (ZKPs), which incur orders-of-magnitude overhead for billion-parameter models, TEEs provide practical cryptographic guarantees with acceptable performance tradeoffs.

Key Takeaways

Close the verification gap with hardware attestation

1

Model substitution is economically motivated and practically undetectable via software.

The profit incentive to swap cheaper models is real, and black-box APIs offer no transparency.

2

FP8 quantization barely affects benchmarks — the signal is buried in statistical noise.

Score differences often fall within error bars, making statistical separation infeasible.

3

Inference nondeterminism kills log-probability fingerprinting.

Software versions and GPU hardware produce divergent token-level probabilities.

4

Randomized substitution defeats statistical tests at query budgets that matter.

At low substitution rates, detection requires millions of queries — economically infeasible.

5

TEEs are the only currently viable path to verifiable LLM APIs.

Hardware attestation provides provable, efficient, and robust model integrity guarantees.



Call to Action: The community should prioritize TEE-based attestation standards for commercial LLM APIs.